

Topics Covered:

- Introduction to Linux
- Linux Commands



What is Linux?

The Linux operating system is a powerful and flexible open-source software platform. It acts as the basis for a variety of devices, such as embedded systems, cell phones, servers, and personal computers. Linux, that's well-known for its reliability, safety, and flexibility, allows users to customize and improve their environment to suit specific needs. Linux is case sensitive.

Linux Commands

1. `pwd` → print working directory
2. `ls` → listing of current directory content
3. `cd` → used for navigation (open directory)
4. `cp` → copy file
5. `rm` → remove files
6. `mkdir` → create new folder
7. `rmdir` → delete an empty folder
8. `rm -r` → delete non empty folder
9. `clear` → clear the terminal (shell)
10. `man` → manuals (display the description about any command)
11. `cat` → Display file contents on the terminal

Examples

1. To print the working directory name

```
(kali@kali)-[~]  
$ pwd  
/home/kali
```



2. List the directory content in the working directory

```
(kali㉿kali)-[~]  
$ ls  
Desktop  Documents  Downloads  Music  myvenv  Pictures  Public  Templates  Videos
```

3. To open Downloads directory and show the folders inside.

```
(kali㉿kali)-[~]  
$ cd Downloads  
  
(kali㉿kali)-[~/Downloads]  
$ ls  
test
```

4. To make a copy of folder

```
(kali㉿kali)-[~/Downloads]  
$ cp -r test testcopy  
  
(kali㉿kali)-[~/Downloads]  
$ ls  
test  testcopy
```

5. Remove a folder

```
(kali㉿kali)-[~/Downloads]  
$ rm -r testcopy  
  
(kali㉿kali)-[~/Downloads]  
$ ls  
test
```

6. Make new folder

```
(kali㉿kali)-[~/Downloads]  
$ mkdir myFolder  
  
(kali㉿kali)-[~/Downloads]  
$ ls  
myFolder  test
```

7. To display a file content

```
(kali㉿kali)-[~/Downloads]  
$ cat myfile  
I take cyber security course at Elsewedy university of technology
```



8. To display description of a command

```
(kali㉿kali)-[~]  
$ man ls
```

```
LS(1)                                User Commands                                LS(1)  
  
NAME  
    ls - list directory contents  
  
SYNOPSIS  
    ls [OPTION]... [FILE]...  
  
DESCRIPTION  
    List information about the FILES (the current directory by default). Sort entries alphabetically if none of -cftuvSUX nor --sort is specified.  
    Mandatory arguments to long options are mandatory for short options too.  
  
    -a, --all  
        do not ignore entries starting with .  
  
    -A, --almost-all
```

Exercises

1. Write Linux commands to go to downloads and make a new folder named **myFolder**, then show all folders in downloads. Make a copy of this folder and name it **copyFolder**.
2. Make file manually named **test** in documents directory and write ("I am in a second year at Faculty of Computer science"), then write a Linux command to go to documents and display the content of this file on the shell.
3. Depending on ex. 1 & 2, remove **copyFolder** and **test** file.

